ALL FOUR RESIDUAL CELLS OF THE DIRECTED HAMILTON–WATERLOO PROBLEM AT ORDER 15 EXIST

ABSTRACT. For odd v and $m, n \mid v$, the directed Hamilton–Waterloo problem $\mathrm{HWP}^*(v; m^r, n^s)$ asks for a decomposition of the arc set of the complete symmetric digraph K_v^* into r spanning factors that are disjoint unions of directed m-cycles and s spanning factors that are disjoint unions of directed n-cycles. Yetgin, Odabaşı and Özkan determine $\mathrm{HWP}^*(15; m^r, n^s)$ for $(m, n) \in \{(3, 5), (3, 15), (5, 15)\}$ apart from four cells, which their printed Lemma 4.2 records as possible exceptions: $r \in \{11, 12, 13\}$ when $(m, n) = (3, 5)$, and $r = 13$ when $(m, n) = (3, 15)$. That lemma poses no problem about those cells; it leaves their four values undetermined. We exhibit a factorization in each of the four, which determines those four values and removes that lemma's exception clause at $v = 15$. Each witness is printed here in full, as 14 fixed-point-free permutations of $\mathbb{Z}_{15}$ whose arc sets partition the 210 arcs of K_{15}^*; a reader can check it by hand and needs no code.

1. THE STATEMENT AND THE FOUR CELLS

Throughout, K_v^* denotes the complete symmetric digraph on v vertices: for every unordered pair $\{x, y\}$ both arcs (x, y) and (y, x) are present, so K_{15}^* has $15 \cdot 14 = 210$ arcs. A *factor* is spanning, and $\mathrm{HWP}^*(v; m^r, n^s)$ *has a solution* when the arc set of K_v^* decomposes into r factors, each a disjoint union of directed m-cycles, and s factors, each a disjoint union of directed n-cycles. Counting arcs and vertices forces $r + s = v - 1$, $m \mid v$ and $n \mid v$.

Yetgin, Odabaşı and Özkan [1] prove, for odd v and $(m, n) \in \{(3, 5), (3, 15), (5, 15)\}$, that $\mathrm{HWP}^*(v; m^r, n^s)$ has a solution if and only if $r + s = v - 1$ and $\mathrm{lcm}(m, n) \mid v$, with a short list of possible exceptions, all at $v = 15$. Their $v = 15$ lemma, printed as Lemma 4.2, reads verbatim:

> For nonnegative integers r and s, $\mathrm{HWP}^*(15; m^r, n^s)$ has a solution for $(m, n) \in \{(3, 5), (3, 15), (5, 15)\}$ if and only if $r + s = 14$ except possibly for $r \in \{11, 12, 13\}$ when $(m, n) = (3, 5)$ and for $r = 13$ when $(m, n) = (3, 15)$.

The quotation is taken from the arXiv e-print of [1] (arXiv:2209.14588v1) and the statement stands unchanged in the version of record; only the macro for HWP^* and the line breaks are ours.

2020 *Mathematics Subject Classification.* 05C70, 05C20, 05B15.

Key words and phrases. Hamilton–Waterloo problem, complete symmetric digraph, directed 2-factorization, sharply transitive set, Latin square.

Since $r + s = 14$, the exception clause names exactly four cells, written below as quadruples (m, n, r, s):

$$(3, 5, 11, 3), \qquad (3, 5, 12, 2), \qquad (3, 5, 13, 1), \qquad (3, 15, 13, 1).$$

Each satisfies the necessary conditions, so the point left undetermined in each cell is sufficiency: whether a witness exists. The source records the four as possible exceptions and poses no problem about them; the theorem below determines the four values it leaves undetermined.

Theorem 1. $\mathrm{HWP}^*(15; 3^{11}, 5^3)$, $\mathrm{HWP}^*(15; 3^{12}, 5^2)$, $\mathrm{HWP}^*(15; 3^{13}, 5^1)$ *and* $\mathrm{HWP}^*(15; 3^{13}, 15^1)$ *all have solutions; a witness for each is printed in Section* 3. *Consequently the clause "except possibly for $r \in \{11, 12, 13\}$ when $(m, n) = (3, 5)$ and for $r = 13$ when $(m, n) = (3, 15)$" may be deleted from printed Lemma* 4.2 *of* [1].

2. The criterion used to check the witnesses

Lemma 2. *Let $m, n \mid 15$ and $r + s = 14$. A solution of* $\mathrm{HWP}^*(15; m^r, n^s)$ *is the same thing as a list $\sigma_1, \ldots, \sigma_{14}$ of fixed-point-free permutations of $\mathbb{Z}_{15}$ such that*

- (i) *each σ_i has all of its cycles of one and the same length, that length being m for r of the indices and n for the remaining s; and*
- (ii) *the* 210 *arcs $(x, \sigma_i(x))$, $x \in \mathbb{Z}_{15}$, $1 \le i \le 14$, are pairwise distinct.*

Equivalently: the 15×15 array whose row 0 is the identity and whose row i is σ_i is a Latin square of order 15.

Proof. A factor of K_{15}^* that is a disjoint union of directed cycles is spanning and has in-degree and out-degree 1 at every vertex, hence is the functional digraph $\{(x, \sigma(x))\}$ of a permutation σ with no fixed point, and its cycle lengths are the cycle lengths of σ; conversely a fixed-point-free permutation all of whose cycles have length m gives a factor consisting of $15/m$ directed m-cycles. Fourteen such factors contribute $14 \cdot 15 = 210$ arcs, which is exactly the number of arcs of K_{15}^*, so they are pairwise arc-disjoint if and only if their union is the whole arc set; that is (ii). For the reformulation, (ii) says that for each fixed x the 14 values $\sigma_1(x), \ldots, \sigma_{14}(x)$ are pairwise distinct, and each differs from x because the σ_i are fixed-point-free; together with the entry x in row 0 they therefore exhaust $\mathbb{Z}_{15}$, i.e. column x of the array is a permutation of $\mathbb{Z}_{15}$. The rows are permutations because the σ_i are. Conversely, in a Latin square of order 15 whose row 0 is the identity, column x already contains the symbol x in row 0, so no other row can have $\sigma_i(x) = x$. □

So the check on each printed witness below is three lines:

- (a) each printed factor is a set of disjoint cycles whose entries are all of $\mathbb{Z}_{15}$, with every cycle length equal to the printed label;
- (b) no cycle has length 1, so each factor is fixed-point-free; and

(c) the 210 arcs are pairwise distinct — equivalently, adjoining the identity, every column of the resulting array is a permutation of $\mathbb{Z}_{15}$.

3. The four witnesses

The vertex set is $\mathbb{Z}_{15}$. Each factor is printed as the permutation σ_i in cycle notation, with its uniform cycle type as a label; its arc set is $\{(x, \sigma_i(x)) : x \in \mathbb{Z}_{15}\}$, i.e. each printed cycle $(a_1, a_2, \ldots, a_k)$ contributes the arcs $a_1 \to a_2 \to \cdots \to a_k \to a_1$, in that direction. In the last listing F_{14} is a single directed cycle through all 15 vertices, as $n = 15$ requires.

```
HWP*(15; 3^11, 5^3)
F1   3^5   (0,1,2)(3,4,5)(6,7,8)(9,10,11)(12,13,14)
F2   3^5   (0,2,5)(1,6,13)(3,8,11)(4,7,12)(9,14,10)
F3   3^5   (0,3,9)(1,14,4)(2,7,6)(5,13,11)(8,12,10)
F4   3^5   (0,4,12)(1,11,6)(2,8,9)(3,5,14)(7,10,13)
F5   3^5   (0,5,11)(1,3,12)(2,6,4)(7,9,8)(10,14,13)
F6   3^5   (0,6,8)(1,7,5)(2,4,14)(3,13,9)(10,12,11)
F7   3^5   (0,7,14)(1,13,3)(2,11,8)(4,6,10)(5,9,12)
F8   3^5   (0,8,13)(1,4,9)(2,10,5)(3,6,12)(7,11,14)
F9   3^5   (0,9,7)(1,10,2)(3,14,8)(4,11,13)(5,12,6)
F10  3^5   (0,10,6)(1,5,8)(2,9,13)(3,7,4)(11,12,14)
F11  3^5   (0,11,1)(2,13,12)(3,10,7)(4,8,5)(6,14,9)
F12  5^3   (0,12,7,2,3)(1,8,14,5,10)(4,13,6,9,11)
F13  5^3   (0,13,8,4,10)(1,12,9,5,7)(2,14,6,3,11)
F14  5^3   (0,14,1,9,4)(2,12,8,10,3)(5,6,11,7,13)

HWP*(15; 3^12, 5^2)
F1   3^5   (0,1,2)(3,4,5)(6,7,8)(9,10,11)(12,13,14)
F2   3^5   (0,2,7)(1,3,14)(4,12,9)(5,6,10)(8,11,13)
F3   3^5   (0,3,6)(1,9,8)(2,4,13)(5,7,14)(10,12,11)
F4   3^5   (0,4,9)(1,10,2)(3,11,6)(5,8,13)(7,12,14)
F5   3^5   (0,5,10)(1,7,11)(2,12,3)(4,14,8)(6,13,9)
F6   3^5   (0,7,1)(2,8,9)(3,5,13)(4,11,12)(6,14,10)
F7   3^5   (0,8,14)(1,11,4)(2,6,5)(3,9,7)(10,13,12)
F8   3^5   (0,9,13)(1,4,6)(2,11,8)(3,10,14)(5,12,7)
F9   3^5   (0,10,4)(1,8,12)(2,13,6)(3,7,9)(5,14,11)
F10  3^5   (0,11,3)(1,14,13)(2,5,4)(6,9,12)(7,10,8)
F11  3^5   (0,12,8)(1,5,9)(2,10,7)(3,13,4)(6,11,14)
F12  3^5   (0,13,11)(1,12,5)(2,9,14)(3,8,10)(4,7,6)
F13  5^3   (0,6,8,3,12)(1,13,7,4,10)(2,14,9,5,11)
F14  5^3   (0,14,4,8,5)(1,6,12,2,3)(7,13,10,9,11)

HWP*(15; 3^13, 5^1)
F1   3^5   (0,1,2)(3,4,5)(6,7,8)(9,10,11)(12,13,14)
F2   3^5   (0,2,4)(1,9,7)(3,11,14)(5,6,13)(8,10,12)
F3   3^5   (0,4,8)(1,6,2)(3,13,9)(5,7,12)(10,14,11)
F4   3^5   (0,5,10)(1,12,6)(2,9,13)(3,14,4)(7,11,8)
F5   3^5   (0,6,5)(1,14,8)(2,11,3)(4,9,12)(7,13,10)
F6   3^5   (0,7,3)(1,5,12)(2,10,9)(4,13,11)(6,8,14)
F7   3^5   (0,8,9)(1,13,4)(2,5,11)(3,12,10)(6,14,7)
F8   3^5   (0,9,1)(2,14,10)(3,5,8)(4,12,7)(6,11,13)
F9   3^5   (0,10,6)(1,11,5)(2,12,14)(3,8,13)(4,7,9)
```

```
F10  3^5   (0,11,7)(1,10,13)(2,8,4)(3,6,12)(5,14,9)
F11  3^5   (0,12,11)(1,7,14)(2,13,8)(3,9,6)(4,10,5)
F12  3^5   (0,13,12)(1,8,11)(2,3,7)(4,6,10)(5,9,14)
F13  3^5   (0,14,13)(1,3,10)(2,7,5)(4,11,6)(8,12,9)
F14  5^3   (0,3,1,4,14)(2,6,9,11,12)(5,13,7,10,8)

HWP*(15; 3^13, 15^1)
F1   3^5   (0,1,2)(3,4,5)(6,7,8)(9,10,11)(12,13,14)
F2   3^5   (0,2,9)(1,7,4)(3,13,10)(5,14,11)(6,8,12)
F3   3^5   (0,4,11)(1,12,3)(2,7,14)(5,8,9)(6,10,13)
F4   3^5   (0,5,13)(1,8,10)(2,3,11)(4,6,14)(7,9,12)
F5   3^5   (0,6,12)(1,11,13)(2,4,10)(3,14,8)(5,9,7)
F6   3^5   (0,7,3)(1,9,14)(2,10,8)(4,12,11)(5,6,13)
F7   3^5   (0,8,1)(2,5,12)(3,10,6)(4,13,9)(7,11,14)
F8   3^5   (0,9,6)(1,5,7)(2,11,3)(4,8,13)(10,12,14)
F9   3^5   (0,10,14)(1,3,12)(2,13,7)(4,9,8)(5,11,6)
F10  3^5   (0,11,7)(1,10,5)(2,6,4)(3,8,14)(9,13,12)
F11  3^5   (0,12,4)(1,14,6)(2,8,5)(3,7,13)(9,11,10)
F12  3^5   (0,13,8)(1,6,11)(2,14,9)(3,5,4)(7,12,10)
F13  3^5   (0,14,5)(1,13,2)(3,6,9)(4,7,10)(8,11,12)
F14  15^1  (0,3,9,1,4,14,13,11,8,7,6,2,12,5,10)
```

Proof of Theorem 1. Each of the four listings satisfies (a), (b) and (c) above, as one verifies directly from the printed cycles; by Lemma 2 each listing is then a solution of the cell named in its header. The accompanying program `verify.py` redoes the same check mechanically on the same printed text (Section 4). □

4. Scope

Combining the four witnesses with the cases already proved in [1] — all of $(m, n) = (5, 15)$, and every $(3, 5)$ and $(3, 15)$ case outside $r \in \{11, 12, 13\}$ and $r = 13$ respectively — printed Lemma 4.2 becomes: for nonnegative r, s, $\mathrm{HWP}^*(15; m^r, n^s)$ has a solution for $(m, n) \in \{(3, 5), (3, 15), (5, 15)\}$ if and only if $r + s = 14$. Those other cases are the authors', and were not re-verified here; what is verified here is the four listings above. Nothing outside $v = 15$ is constructed or claimed. In particular Lemma 4.1 of [1], which lifts a complete list of solutions at one odd order to larger odd orders, is the authors' own; it is neither verified nor used here.

The four witnesses were found by a constraint solver; nothing in Theorem 1 depends on the solver, a re-run would return different objects, and the objects printed above are the claim.

The program `verify.py` that accompanies this note calls no solver: it parses the listing of Section 3 out of its own source — the same characters as are printed above — and re-derives, for each cell, the necessary conditions, the factor count, that every factor is a fixed-point-free permutation of $\mathbb{Z}_{15}$ whose cycle type matches its printed label, that the census of labels matches the cell, that the 210 arcs are exactly the arcs of K_{15}^* (as a set identity: none missing and none repeated), and that the array obtained by adjoining the identity is a Latin square. The four cells are re-derived from the wording of the exception clause rather than copied from the listing. No non-existence, census or exhaustion claim is made or checked anywhere: every verdict is the existence of an exhibited object.

Nothing beyond the four factorizations printed above is claimed.

References

[1] F. Yetgin, U. Odabaşı, and S. Özkan, *On the directed Hamilton–Waterloo problem with two cycle sizes*, Contrib. Discrete Math. **20** (2025), no. 1, 74–94. doi:10.55016/ojs/cdm.v20i1.75051; e-print arXiv:2209.14588. Statement numbers cited here are those of the version of record.